(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: BS-112 ***Subject: Applied Mathematics-II***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No. 1 which is compulsory. Select one question from each unit.

Q1. (a) *Find all the point at which the following mapping is not Conformal* $w = \frac{z+\frac{1}{2}}{4z^2+2}$. **(2.5)**

(b) *Split the real and imaginary part of* i^i. **(2.5)**

(c). *Find the Laplace transform of* t^2e^{-2t}. **(2.5)**

(d) *Using half range sine series of function* $f(x) = 1$ *for* $0 < x < \pi$,

prove that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \ldots\ldots = \frac{\pi^2}{8}$ **(2.5)**

(e) *Taylor series expansion of* $\frac{1}{z-2}$ *in* $|z| < 1$ *is*..... **(2.5)**

(f) *Classify the type of PDE:* $\frac{\partial^2 u}{\partial x^2} + 2\frac{\partial^2 u}{\partial x \partial y} + 5\frac{\partial^2 u}{\partial y^2} = 0$, *whether it is parabolic, ellipticor hyperbolic?* **(2.5)**

UNIT-I

Q 2. (a) *Find all the values of* $(-1 + \sqrt{3}\, i)^{\frac{3}{2}}$. **(7)**

(b)*Verify that the function* $u(x, y) = x^3y - xy^3$ *is harmonic and find the harmonic conjugate of* $u(x, y)$ *to express the function* $f(z) = u + iv$ *as an analytic function.* **(8)**

Q 3. (a)*Evaluate the integral* $\oint_C \frac{e^{2z}}{(z-1)(z-2)} dz$ *along the curve* C, *where* C *is a circle* $|z| = 3$. **(7)**

(b) *Integrate the function* $f(z) = \bar{z}$ *along the curve* C, *where* C *is the square with vertices* $z = 0, 2, 2i, 2 + 2i$ **(8)**

UNIT-II

Q 4.(a) *Find the bilinear transformation or Mobious transformation which maps* $1, i, 1$ *of the* $z-$*plane onto* $1, i, -1$ *of the* $w-$*plane respectively. Also find the fixed points or invariant points.* **(8)**

P.T.O.